1) **Write a ‘C’ program to accept a number and check if it is positive, negative or zero.**

Ans ->

```
#include <stdio.h>
int main()
{
   int num;

   printf("Enter a number: ");
   scanf("%d", &num);

   if (num > 0)
      printf("Positive number");
   else if (num < 0)
      printf("Negative number");
   else
      printf("Zero");

   return 0;
}
```

I/P- Enter a number: 10
Positive number

Enter a number: -5
Negative number

Enter a number: 0
Zero

2) **Write a ‘C’ program to display multiplication table from 1 to ‘n’ integer with 10 rows.**

Ans -

```
#include <stdio.h>
int main()
{
   int n, i, j;

   printf("Enter a number: ");
   scanf("%d", &n);

   for (i = 1; i <= n; i++)
   {
      for (j = 1; j <= 10; j++)
      {
         printf("%d x %d = %d\n", i, j, i * j);
      }
      printf("\n");
   }
```

```
    return 0;
}
```

O/P - Enter a number: 2

1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
...
1 x 10 = 10

2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
...
2 x 10 = 20

3) **Write a 'C' program to accept 'n' integers in an array and display in sorted manner using function.**

Ans ->

```
#include <stdio.h>
void sort(int a[], int n)
{
    int i, j, temp;

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (a[i] > a[j])
            {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }
}

int main()
{
    int a[100], n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
```

```
    sort(a, n);

    printf("Sorted array:\n");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

o/p-Enter number of elements: 5
Enter elements:
40 10 30 20 50
Sorted array:
10 20 30 40 50

4) **Write a 'C' program to perform all arithmetic operations on two integers.**

Ans -

```
#include <stdio.h>
int main()
{
    int a, b;

    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);

    printf("Addition = %d\n", a + b);
    printf("Subtraction = %d\n", a - b);
    printf("Multiplication = %d\n", a * b);

    if (b != 0)
    {
        printf("Division = %d\n", a / b);
        printf("Modulus = %d\n", a % b);
    }
    else
    {
        printf("Division and Modulus not possible");
    }

    return 0;
}
```

o/p-> Enter two integers: 10 3
Addition = 13
Subtraction = 7
Multiplication = 30
Division = 3
Modulus = 1

5) **Write a ‘C’ program to accept a number from the user and display it in words. Input = 89, output = Eight Nine**

Ans ->

```
#include <stdio.h>
int main()
{
   int n, digit, rev = 0;

   printf("Enter a number: ");
   scanf("%d", &n);

   while (n > 0)
   {
      rev = rev * 10 + n % 10;
      n = n / 10;
   }

   while (rev > 0)
   {
      digit = rev % 10;

      switch (digit)
      {
         case 0: printf("Zero "); break;
         case 1: printf("One "); break;
         case 2: printf("Two "); break;
         case 3: printf("Three "); break;
         case 4: printf("Four "); break;
         case 5: printf("Five "); break;
         case 6: printf("Six "); break;
         case 7: printf("Seven "); break;
         case 8: printf("Eight "); break;
         case 9: printf("Nine "); break;
      }

      rev = rev / 10;
   }

   return 0;
}
```

O/P->Enter a number: 89

Eight Nine

6) **Write a ‘C’ program to accept two matrices of size m x n and calculate Addition and Subtraction of matrices using with separate functions.**

Ans ->

```
#include <stdio.h>
void add(int a[10][10], int b[10][10], int m, int n)
{
```

```
    int i, j;

    printf("Addition:\n");
    for (i = 0; i < m; i++)
    {
        for (j = 0; j < n; j++)
            printf("%d ", a[i][j] + b[i][j]);
        printf("\n");
    }
}

void sub(int a[10][10], int b[10][10], int m, int n)
{
    int i, j;

    printf("Subtraction:\n");
    for (i = 0; i < m; i++)
    {
        for (j = 0; j < n; j++)
            printf("%d ", a[i][j] - b[i][j]);
        printf("\n");
    }
}

int main()
{
    int a[10][10], b[10][10];
    int m, n, i, j;

    printf("Enter rows and columns: ");
    scanf("%d%d", &m, &n);

    printf("Enter first matrix:\n");
    for (i = 0; i < m; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &a[i][j]);

    printf("Enter second matrix:\n");
    for (i = 0; i < m; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &b[i][j]);

    add(a, b, m, n);
    sub(a, b, m, n);

    return 0;
}
```

o/p- Enter rows and columns: 2 2

Enter first matrix:
1 2
3 4
Enter second matrix:
5 6
7 8

Addition:
6 8
10 12

Subtraction:
-4 -4
-4 -4

7) **Write a 'C' program to Accept dimensions of a cylinder and print the surface area and volume. (hint : surface area = 2π r2 + 2π r h, volume = π r2 h)**

Ans -

```
#include <stdio.h>
#define PI 3.14

int main()
{
   float r, h, surface_area, volume;

   printf("Enter radius and height: ");
   scanf("%f %f", &r, &h);

   surface_area = 2 * PI * r * r + 2 * PI * r * h;
   volume = PI * r * r * h;

   printf("Surface Area = %.2f\n", surface_area);
   printf("Volume = %.2f\n", volume);

   return 0;
}
```

O/P- Enter radius and height: 5 10
Surface Area = 471.00
Volume = 785.00

8) **Write a 'C' program to accept two matrices of size m x n and display multiplication of matrices.**

Ans -

```
#include <stdio.h>
int main()
{
   int a[10][10], b[10][10], c[10][10];
```

```
    int m, n, i, j, k;

    printf("Enter rows and columns: ");
    scanf("%d %d", &m, &n);

    printf("Enter first matrix:\n");
    for(i = 0; i < m; i++)
    {
        for(j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    }

    printf("Enter second matrix:\n");
    for(i = 0; i < m; i++)
    {
        for(j = 0; j < n; j++)
            scanf("%d", &b[i][j]);
    }

    for(i = 0; i < m; i++)
    {
        for(j = 0; j < n; j++)
        {
            c[i][j] = 0;
            for(k = 0; k < n; k++)
                c[i][j] += a[i][k] * b[k][j];
        }
    }

    printf("Multiplication of matrices:\n");
    for(i = 0; i < m; i++)
    {
        for(j = 0; j < n; j++)
            printf("%d ", c[i][j]);
        printf("\n");
    }

    return 0;
}
```

O/p - Enter rows and columns: 2 2
Enter first matrix:
1 2
3 4
Enter second matrix:
5 6
7 8

Multiplication of matrices:

19 22
43 50

9)Write a ‘C’ program to Accept two integers x and y and calculate the sum of all integers between x and y using call by value.
Ans -

```
 #include <stdio.h>
int sum(int x, int y)
{
   int i, s = 0;

   for(i = x; i <= y; i++)
   {
      s = s + i;
   }

   return s;
}

int main()
{
   int x, y, result;

   printf("Enter two integers: ");
   scanf("%d %d", &x, &y);

   result = sum(x, y);

   printf("Sum = %d", result);

   return 0;
}
```

O/p -Enter two integers: 1 5
Sum = 15